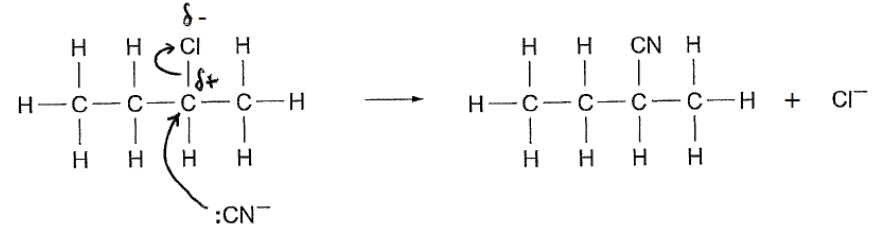
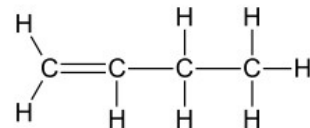
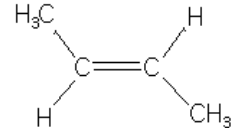
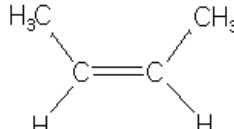


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		 $\delta+$ and $\delta-$ shown on C and Cl (1) arrow from lone pair on CN^- to $\delta+$ C (1) arrow from C—Cl bond to Cl (1) 2-cyanobutane and Cl^- as products (1)	1					
		(ii)		nucleophilic substitution	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		 but-1-ene (1)  E-but-2-ene (1)  Z-but-2-ene (1) award (1) for all formulae with no names		3		3		
		(ii)		elimination	1			1		
	(c)			award (1) for each of the following • CFCs destroy ozone • we are then more exposed to UV radiation / greater risk of skin cancer • (CFCs) produce •Cl / chlorine radical / HFCs do not produce •Cl • •Cl / chlorine radical starts chain reaction • HFCs contain C—H / C—F bonds and these are too strong to be broken by UV light	5			5		
				Question 10 total	9	5	0	14	0	0